CLIENT	**:**	**OBEROI REALTIY**
SITE	**:**	**MAXIMA, MUMBAI**
LOCATION	**:**	**MUMBAI, MHARASHTRA**
NO.	**:**	**-**
DOCUMENT NO.	**:**	**DBR-R00**

R00	04-03-2023	Design report	R.H.B	H.A.B	H.A.B
Rev No.	**Date**	**Description**	**Prepared By**	**Checked By**	**Approved By**

TABLE OF CONTENTS

Sr. No	Contents	Page No.
1	Design Philosophy	3
2	Design Codes and Standards	4
3	Material Specifications Construction	4
4	Load Calculations	5
4.1	Dead Load	5
4.2	Live Load	6
4.3	Wind Load Calculations	7
5	Deflection Check	8
6	Primary Loads and Load Combinations	8

1. Design Philosophy

This document pertains to structural design of **Maxima roof**, Mumbai, Maharashtra at 28th floor. The overall size of the frame is 4.737 x 4.60 x 0.350m high. The main sliding door frame is of Aluminium 6063-T6 material. The wind load is worked out from IS 875 Part III. The various DL and LL acting at different points on frame are transferred to the nearest columns through the grid of beams.

The complete frame is modelled as 3D frame and analysed/designed for worst combination of loads through STAAD.pro. Input and output files are enclosed which are self-explanatory. From the STAAD output, the support reactions for various primary load cases are tabulated and the base plate, shear keys and foundation bolts are designed accordingly. Foundation design shall be done using appropriate load combinations for reactions given in this document.

2. DESIGN CODES AND STANDARDS

Code / Standard	Applicability	Remark
IS800-2007	Design of steel structures	--
IS 875-Part III	For wind load calculation	--

3. MATERIAL SPECIFICATION FOR CONSTRUCTION

Sr.no	Material	Specification	Minimum Strength
1	Aluminium	Aluminium 6063-T6	Fy = 214 N/mm^2

4. LOAD CALCULATIONS

4.1 DEAD LOAD

SELFWEIGHT FACTOR = 1

Dead load for glass/sheet assumed = 5 Kg/m^2 i.e. 0.05 kN/m^2

4.2 LIVE LOAD

Live load assumed = 0.75 kN/m^2

4.3 WIND LOAD

For wind load calculation, following IS code is referred: - “IS 875 Part III”

DESIGN WIND PRESSURE CALCULATION				
Sr no:	**Description**	**Value**	**Unit**	**Remarks**
1	V_b Basic Wind Speed	44	m/s	As per Apendix A (IS-875 Part 3)
2	K1 (risk coefficient)	1		For Design life span of 50 yrs. As per IS 875 Part III (table-1)
3	K2 (Terrain roughness and height factor) select terrain cat.	Catagory2		
	K2 for height 10m	10	1	As per IS 875 Part III (table-2), Cat-2, As per Ht
	K2 for height 15m	15	1.05	
	K2 for height 20m	20	1.07	
	K2 for height 30m	30	1.12	
	K2 for height 50m	50	1.17	
	K2 for height 100m	100	1.24	
4	K3 (topography factor)	1		Clause 6.3.3 IS 875 part III
5	K4 (Importance factor for Cyclonic region.)	1.15		Clause 6.3.4 IS 875 part III
6	K_a (Area Averaging factor)	0.89		As per IS 875 Part III (table-4)
7	K_d (Wind Directionality Factor)	0.9		Clause 7.2.1 IS 875 part III
8	K_c (Combination Factor)	0.9		Clause 7.3.3.13 IS 875 part III
	Vz=	Vb*K1*K2*K3*K4		
	Vz (For 10m ht)	50.6	**m/s**	Vz= Vb*K1*K2*K3*K4
	Vz (For 15m ht)	53.13	**m/s**	
	Vz (For 20m ht)	54.142	**m/s**	
	Vz (For 30m ht)	56.672	**m/s**	
	Vz (For 50m ht)	59.202	**m/s**	
	Vz (For 100m ht)	62.744	**m/s**	
	Pz (For 10m ht)	1.54	kN/m2	$Pz = 0.6(Vz)^2$
	Pz (For 15m ht)	1.69	kN/m2	
	Pz (For 20m ht)	1.76	kN/m2	
	Pz (For 30m ht)	1.93	kN/m2	
	Pz (For 50m ht)	2.10	kN/m2	
	Pz (For 100m ht)	2.36	kN/m2	
	Ka*Kd*Kc =	0.72		**Use this value**
	Hence Use pd (For 10m Ht)	1.11	kN/m2	
	Hence Use pd (For 15m Ht)	1.23	kN/m2	
	Hence Use pd (For 20m Ht)	1.27	kN/m2	
	Hence Use pd (For 30m Ht)	1.40	kN/m2	
	Hence Use pd (For 50m Ht)	1.52	kN/m2	
	Hence Use pd (For 100m Ht)	1.71	kN/m2	
		1.11	10	
		1.23	20	
		1.27	25	
		1.40	30	
		1.52	50	
		1.71	100	

	Vz	**44**	**m/s**	
	Pz =	1.16	kN/m2	$Pz = 0.6(Vz)^2$
	Pd1 =	0.75	kN/m2	Pd1=Pz*Ka*Kd*Kc
	or Pd2 =	0.81	kN/m2	Pd2=0.7*Pz
	Hence Use pd =	0.81	kN/m2	Max of Pd1 and Pd2
	Coefficeint for Pressure Case=	0.6		
	Coefficeint for Suction Case=	0.3		
	Span =	1		
	UDL for Pressure case =	0.5	kN/m	

5. Deflection check at building top in Z direction

Refer node no. –5 of staad file for all load cases

Max. Deflection in Z dir. = 16.5 mm < H/250 = 4737/240 = 19.73mm (IS 800-2007 table 6)

.... Hence ok.

6. PRIMARY LOADS & LOAD COMBINATIONS

6.1 PRIMARY LOADS

Load 1 Dead Load (DL)

Load 2 Live Load (LL)

Load 3 Wind Load Pressure (WLX)

Load 4 Wind Load Suction (WLZ)

Load 5 Wind Load Suction (WL-X)

Load 6 Wind Load Suction (WL-Z)

6.2 LOAD COMBINATIONS

Load Combination for strength:-

1.5 DL + 1.5 LL
1.5 DL + 1.5 WLX
1.5 DL + 1.5 WLZ
1.5 DL + 1.5 WL-X
1.5 DL + 1.5 WL-Z

Load Combination for serviceability:-

DL + LL
DL + WLX
DL + WLZ
DL + WL-X
DL + WL-Z
DL + 0.8 WLX
DL + 0.8 WLZ
DL + 0.8 WL-X
DL + 0.8 WL-Z

```
            **************************************************
            *                                                *
            *          STAAD.Pro CONNECT Edition             *
            *          Version  22.02.00.26                  *
            *          Proprietary Program of                *
            *          Bentley Systems, Inc.                 *
            *          Date=    APR  3, 2023                 *
            *          Time=    17:42:14                     *
            *                                                *
            *  Licensed to: only @ ::LAVTeam:: (R)           *
            **************************************************



     1. STAAD SPACE
INPUT FILE: D:\SALT\MISC\MAXIMA\STAAD\MAXIMA (03.04.2023).STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 16-DEC-22
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1 0 84.0002 0; 2 4.73701 84.0002 0; 3 0 84.2102 0; 4 4.73701 84.2102 0
     9. 5 2.3685 84.3502 0; 6 0.590291 84.2714 0; 7 1.18214 84.3151 0
    10. 8 1.775 84.3412 0; 9 4.14672 84.2714 0; 10 3.55487 84.3151 0
    11. 11 2.96201 84.3412 0; 12 0 84.0002 -1.075; 13 4.73701 84.0002 -1.075
    12. 14 0 84.2102 -1.075; 15 4.73701 84.2102 -1.075; 16 2.3685 84.3502 -1.075
    13. 17 0.590291 84.2714 -1.075; 18 1.18214 84.3151 -1.075; 19 1.775 84.3412 -1.075
    14. 20 4.14672 84.2714 -1.075; 21 3.55487 84.3151 -1.075
    15. 22 2.96201 84.3412 -1.075
    16. MEMBER INCIDENCES
    17. 1 1 3; 2 2 4; 3 3 6; 4 6 7; 5 7 8; 6 8 5; 7 4 9; 8 9 10; 9 10 11; 10 11 5
    18. 13 3 14; 14 4 15; 15 5 16; 16 6 17; 17 7 18; 18 8 19; 19 9 20; 20 10 21
    19. 21 11 22; 22 12 14; 23 13 15; 24 14 17; 25 17 18; 26 18 19; 27 19 16; 28 15 20
    20. 29 20 21; 30 21 22; 31 22 16
    21. SUPPORTS
    22. 1 2 12 13 PINNED
    23. MEMBER RELEASE
    24. 13 TO 21 START MY MZ
    25. 13 TO 21 END MY MZ
    26. DEFINE MATERIAL START
    27. ISOTROPIC ALUMINUM
    28. E 6.89473E+07
    29. POISSON 0.33
    30. DENSITY 26.6017
    31. ALPHA 1.28E-05
    32. DAMP 0.03
    33. ISOTROPIC STEEL
    34. E 1.99947E+08
    35. POISSON 0.3
    36. DENSITY 76.8191
    37. ALPHA 6.5E-06
    38. DAMP 0.03
```

```
39. TYPE STEEL
40. STRENGTH RY 1.5 RT 1.2
41. END DEFINE MATERIAL
42. CONSTANTS
43. MATERIAL ALUMINUM ALL
44. MEMBER PROPERTY APLAPOLLOTUBES
45. 13 TO 21 TABLE ST 25X25X2.0SHS
46. MEMBER PROPERTY APLAPOLLOTUBES
47. 1 TO 10 22 TO 31 TABLE ST 60X60X3.2SHS
48. DEFINE WIND LOAD

*** NOTE: If any floor diaphragm is present in the model Wind Load definition
          should be defined after Floor Diaphragm definition. Otherwise wind
          load generation may be unsuccessful during analysis.

49. TYPE 1 WIND 1
50. INT 1.11 1.23 1.27 1.4 1.52 1.71 HEIG  10  20  25  30  50  100
51. LOAD 1 LOADTYPE DEAD  TITLE DL
52. MEMBER LOAD
53. 1 TO 10 22 TO 31 UNI GY -0.025
54. LOAD 2 LOADTYPE DEAD  TITLE LL
55. MEMBER LOAD
56. 3 TO 10 24 TO 31 UNI GY -0.35
57. LOAD 3 LOADTYPE NONE  TITLE WL X
58. WIND LOAD X 1 TYPE 1 XR 0 4.737 YR 84 84.35 ZR -1.075 0
59. MEMBER LOAD
60. 3 TO 10 24 TO 31 UNI GY -0.5
61. LOAD 4 LOADTYPE NONE  TITLE WL Z
62. WIND LOAD Z 1 TYPE 1 XR 0 4.737 YR 84 84.35 ZR -1.075 0
63. MEMBER LOAD
64. 3 TO 10 24 TO 31 UNI GY -0.5
65. LOAD 5 LOADTYPE NONE  TITLE WL-X
66. WIND LOAD X -1 TYPE 1 XR 0 4.737 YR 84 84.35 ZR -1.075 0
67. MEMBER LOAD
68. 3 TO 10 24 TO 31 UNI GY -0.5
69. LOAD 6 LOADTYPE NONE  TITLE WL-Z
70. WIND LOAD Z 1 TYPE 1 XR 0 4.737 YR 84 84.35 ZR -1.075 0
71. MEMBER LOAD
72. 3 TO 10 24 TO 31 UNI GY -0.5
73. LOAD COMB 101 1.5(DL+LL)
74. 1 1.5 2 1.5
75. LOAD COMB 102 1.5(DL+WLX)
76. 1 1.5 3 1.5
77. LOAD COMB 103 1.5(DL+WLZ)
```

STAAD SPACE -- PAGE NO. 3

```
 78. 1 1.5 4 1.5
 79. LOAD COMB 104 1.5(DL+WL-X)
 80. 1 1.5 5 1.5
 81. LOAD COMB 105 1.5(DL+WL-Z)
 82. 1 1.5 6 1.5
 83. LOAD COMB 201 DL + LL
 84. 1 1.0 2 1.0
 85. LOAD COMB 202 DL+WLX
 86. 1 1.0 3 1.0
 87. LOAD COMB 203 DL+WLZ
 88. 4 1.0 1 1.0
 89. LOAD COMB 204 DL+WL-X
 90. 1 1.0 5 1.0
 91. LOAD COMB 205 DL+WL-Z
 92. 6 1.0 1 1.0
 93. LOAD COMB 206 DL+0.8WLX
 94. 1 1.0 3 0.8
 95. LOAD COMB 207 DL+0.8WLZ
 96. 4 0.8 1 1.0
 97. LOAD COMB 208 DL+0.8WL-X
 98. 5 0.8 1 1.0
 99. LOAD COMB 209 DL+0.8WL-Z
100. 6 0.8 1 1.0
101. PERFORM ANALYSIS PRINT ALL


         P R O B L E M   S T A T I S T I C S
         -----------------------------------

  NUMBER OF JOINTS         22  NUMBER OF MEMBERS      29
  NUMBER OF PLATES          0  NUMBER OF SOLIDS        0
  NUMBER OF SURFACES        0  NUMBER OF SUPPORTS      4

        Using 64-bit analysis engine.

        SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

  ORIGINAL/FINAL BAND-WIDTH=    11/     3/     21 DOF
TOTAL      PRIMARY LOAD CASES =     6, TOTAL DEGREES OF FREEDOM =     120
TOTAL LOAD COMBINATION  CASES =    14  SO FAR.
  SIZE OF STIFFNESS MATRIX =        3 DOUBLE  KILO-WORDS
  REQRD/AVAIL. DISK SPACE  =     12.1/ 193183.4 MB
```

STAAD SPACE -- PAGE NO. 4

LOADING 1 LOADTYPE DEAD TITLE DL

MEMBER LOAD - UNIT KN METE

MEMBER	UDL	L1	L2	CON	L	LIN1	LIN2
1	-0.0250 GY	0.00	0.21				
2	-0.0250 GY	0.00	0.21				
3	-0.0250 GY	0.00	0.59				
4	-0.0250 GY	0.00	0.59				
5	-0.0250 GY	0.00	0.59				
6	-0.0250 GY	0.00	0.59				
7	-0.0250 GY	0.00	0.59				
8	-0.0250 GY	0.00	0.59				
9	-0.0250 GY	0.00	0.59				
10	-0.0250 GY	0.00	0.59				
22	-0.0250 GY	0.00	0.21				
23	-0.0250 GY	0.00	0.21				
24	-0.0250 GY	0.00	0.59				
25	-0.0250 GY	0.00	0.59				
26	-0.0250 GY	0.00	0.59				
27	-0.0250 GY	0.00	0.59				
28	-0.0250 GY	0.00	0.59				
29	-0.0250 GY	0.00	0.59				
30	-0.0250 GY	0.00	0.59				
31	-0.0250 GY	0.00	0.59				

LOADING 2 LOADTYPE DEAD TITLE LL

MEMBER LOAD - UNIT KN METE

MEMBER	UDL	L1	L2	CON	L	LIN1	LIN2
3	-0.3500 GY	0.00	0.59				
4	-0.3500 GY	0.00	0.59				
5	-0.3500 GY	0.00	0.59				
6	-0.3500 GY	0.00	0.59				
7	-0.3500 GY	0.00	0.59				
8	-0.3500 GY	0.00	0.59				
9	-0.3500 GY	0.00	0.59				
10	-0.3500 GY	0.00	0.59				
24	-0.3500 GY	0.00	0.59				
25	-0.3500 GY	0.00	0.59				
26	-0.3500 GY	0.00	0.59				
27	-0.3500 GY	0.00	0.59				
28	-0.3500 GY	0.00	0.59				
29	-0.3500 GY	0.00	0.59				
30	-0.3500 GY	0.00	0.59				
31	-0.3500 GY	0.00	0.59				

STAAD SPACE -- PAGE NO. 5

LOADING 3 LOADTYPE NONE TITLE WL X

JOINT LOAD - UNIT KN METE

JOINT	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
1	0.10	0.00	0.00	0.00	0.00	0.00
3	0.12	0.00	0.00	0.00	0.00	0.00
5	0.00	0.00	0.00	0.00	0.00	0.00
6	0.05	0.00	0.00	0.00	0.00	0.00
7	0.03	0.00	0.00	0.00	0.00	0.00
8	0.02	0.00	0.00	0.00	0.00	0.00
12	0.10	0.00	0.00	0.00	0.00	0.00
14	0.12	0.00	0.00	0.00	0.00	0.00
16	0.00	0.00	0.00	0.00	0.00	0.00
17	0.05	0.00	0.00	0.00	0.00	0.00
18	0.03	0.00	0.00	0.00	0.00	0.00
19	0.02	0.00	0.00	0.00	0.00	0.00

MEMBER LOAD - UNIT KN METE

MEMBER	UDL	L1	L2	CON	L	LIN1	LIN2
3	-0.5000 GY	0.00	0.59				
4	-0.5000 GY	0.00	0.59				
5	-0.5000 GY	0.00	0.59				
6	-0.5000 GY	0.00	0.59				
7	-0.5000 GY	0.00	0.59				
8	-0.5000 GY	0.00	0.59				
9	-0.5000 GY	0.00	0.59				
10	-0.5000 GY	0.00	0.59				
24	-0.5000 GY	0.00	0.59				
25	-0.5000 GY	0.00	0.59				
26	-0.5000 GY	0.00	0.59				
27	-0.5000 GY	0.00	0.59				
28	-0.5000 GY	0.00	0.59				
29	-0.5000 GY	0.00	0.59				
30	-0.5000 GY	0.00	0.59				
31	-0.5000 GY	0.00	0.59				

LOADING 4 LOADTYPE NONE TITLE WL Z

JOINT LOAD - UNIT KN METE

JOINT	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
12	0.00	0.00	0.52	0.00	0.00	0.00
13	0.00	0.00	0.52	0.00	0.00	0.00

14	0.00	0.00	0.29	0.00	0.00	0.00
15	0.00	0.00	0.29	0.00	0.00	0.00
16	0.00	0.00	0.10	0.00	0.00	0.00
17	0.00	0.00	0.14	0.00	0.00	0.00
18	0.00	0.00	0.12	0.00	0.00	0.00
19	0.00	0.00	0.10	0.00	0.00	0.00
20	0.00	0.00	0.14	0.00	0.00	0.00
21	0.00	0.00	0.12	0.00	0.00	0.00
22	0.00	0.00	0.10	0.00	0.00	0.00

MEMBER LOAD - UNIT KN METE

MEMBER	UDL	L1	L2	CON	L	LIN1	LIN2
3	-0.5000 GY	0.00	0.59				
4	-0.5000 GY	0.00	0.59				
5	-0.5000 GY	0.00	0.59				
6	-0.5000 GY	0.00	0.59				
7	-0.5000 GY	0.00	0.59				
8	-0.5000 GY	0.00	0.59				
9	-0.5000 GY	0.00	0.59				
10	-0.5000 GY	0.00	0.59				
24	-0.5000 GY	0.00	0.59				
25	-0.5000 GY	0.00	0.59				
26	-0.5000 GY	0.00	0.59				
27	-0.5000 GY	0.00	0.59				
28	-0.5000 GY	0.00	0.59				
29	-0.5000 GY	0.00	0.59				
30	-0.5000 GY	0.00	0.59				
31	-0.5000 GY	0.00	0.59				

LOADING 5 LOADTYPE NONE TITLE WL-X

JOINT LOAD - UNIT KN METE

JOINT	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
2	-0.10	0.00	0.00	0.00	0.00	0.00
4	-0.12	0.00	0.00	0.00	0.00	0.00
5	-0.00	0.00	0.00	0.00	0.00	0.00
9	-0.05	0.00	0.00	0.00	0.00	0.00
10	-0.03	0.00	0.00	0.00	0.00	0.00
11	-0.02	0.00	0.00	0.00	0.00	0.00
13	-0.10	0.00	0.00	0.00	0.00	0.00
15	-0.12	0.00	0.00	0.00	0.00	0.00
16	-0.00	0.00	0.00	0.00	0.00	0.00
20	-0.05	0.00	0.00	0.00	0.00	0.00
21	-0.03	0.00	0.00	0.00	0.00	0.00
22	-0.02	0.00	0.00	0.00	0.00	0.00

STAAD SPACE -- PAGE NO. 7

MEMBER LOAD - UNIT KN METE

MEMBER	UDL	L1	L2	CON	L	LIN1	LIN2
3	-0.5000 GY	0.00	0.59				
4	-0.5000 GY	0.00	0.59				
5	-0.5000 GY	0.00	0.59				
6	-0.5000 GY	0.00	0.59				
7	-0.5000 GY	0.00	0.59				
8	-0.5000 GY	0.00	0.59				
9	-0.5000 GY	0.00	0.59				
10	-0.5000 GY	0.00	0.59				
24	-0.5000 GY	0.00	0.59				
25	-0.5000 GY	0.00	0.59				
26	-0.5000 GY	0.00	0.59				
27	-0.5000 GY	0.00	0.59				
28	-0.5000 GY	0.00	0.59				
29	-0.5000 GY	0.00	0.59				
30	-0.5000 GY	0.00	0.59				
31	-0.5000 GY	0.00	0.59				

LOADING 6 LOADTYPE NONE TITLE WL-Z

JOINT LOAD - UNIT KN METE

JOINT	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
12	0.00	0.00	0.52	0.00	0.00	0.00
13	0.00	0.00	0.52	0.00	0.00	0.00
14	0.00	0.00	0.29	0.00	0.00	0.00
15	0.00	0.00	0.29	0.00	0.00	0.00
16	0.00	0.00	0.10	0.00	0.00	0.00
17	0.00	0.00	0.14	0.00	0.00	0.00
18	0.00	0.00	0.12	0.00	0.00	0.00
19	0.00	0.00	0.10	0.00	0.00	0.00
20	0.00	0.00	0.14	0.00	0.00	0.00
21	0.00	0.00	0.12	0.00	0.00	0.00
22	0.00	0.00	0.10	0.00	0.00	0.00

MEMBER LOAD - UNIT KN METE

MEMBER	UDL	L1	L2	CON	L	LIN1	LIN2
3	-0.5000 GY	0.00	0.59				
4	-0.5000 GY	0.00	0.59				
5	-0.5000 GY	0.00	0.59				
6	-0.5000 GY	0.00	0.59				
7	-0.5000 GY	0.00	0.59				
8	-0.5000 GY	0.00	0.59				
9	-0.5000 GY	0.00	0.59				

STAAD SPACE -- PAGE NO. 8

10	-0.5000 GY	0.00	0.59
24	-0.5000 GY	0.00	0.59
25	-0.5000 GY	0.00	0.59
26	-0.5000 GY	0.00	0.59
27	-0.5000 GY	0.00	0.59
28	-0.5000 GY	0.00	0.59
29	-0.5000 GY	0.00	0.59
30	-0.5000 GY	0.00	0.59
31	-0.5000 GY	0.00	0.59

***WARNING - INSTABILITY AT JOINT 2 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.9718504E+03 L-MATRIX DIAG= -5.4569682E-12 EQN NO 118
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

FOR LOADING - 1
APPLIED JOINT EQUIVALENT LOADS

JOINT	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
1	0.00000E+00	-2.62503E-03	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
2	0.00000E+00	-2.62503E-03	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
3	-7.88979E-11	-1.00432E-02	0.00000E+00	0.00000E+00	0.00000E+00	-7.29815E-04
4	-6.67632E-11	-1.00432E-02	0.00000E+00	0.00000E+00	0.00000E+00	7.29812E-04
5	9.75866E-12	-1.48393E-02	0.00000E+00	0.00000E+00	0.00000E+00	-2.49910E-08
6	-1.51578E-10	-1.48364E-02	0.00000E+00	0.00000E+00	0.00000E+00	-1.93252E-06
7	-5.06120E-12	-1.48362E-02	0.00000E+00	0.00000E+00	0.00000E+00	-1.21798E-06
8	4.60564E-11	-1.48375E-02	0.00000E+00	0.00000E+00	0.00000E+00	-9.56866E-07
9	1.58720E-10	-1.48364E-02	0.00000E+00	0.00000E+00	0.00000E+00	1.93851E-06
10	-7.11797E-11	-1.48362E-02	0.00000E+00	0.00000E+00	0.00000E+00	1.21414E-06
11	-2.18820E-11	-1.48377E-02	0.00000E+00	0.00000E+00	0.00000E+00	9.82647E-07
12	0.00000E+00	-2.62503E-03	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
13	0.00000E+00	-2.62503E-03	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
14	-7.88979E-11	-1.00432E-02	0.00000E+00	0.00000E+00	0.00000E+00	-7.29815E-04
15	-6.67632E-11	-1.00432E-02	0.00000E+00	0.00000E+00	0.00000E+00	7.29812E-04
16	9.75866E-12	-1.48393E-02	0.00000E+00	0.00000E+00	0.00000E+00	-2.49910E-08
17	-1.51578E-10	-1.48364E-02	0.00000E+00	0.00000E+00	0.00000E+00	-1.93252E-06
18	-5.06120E-12	-1.48362E-02	0.00000E+00	0.00000E+00	0.00000E+00	-1.21798E-06
19	4.60564E-11	-1.48375E-02	0.00000E+00	0.00000E+00	0.00000E+00	-9.56866E-07
20	1.58720E-10	-1.48364E-02	0.00000E+00	0.00000E+00	0.00000E+00	1.93851E-06
21	-7.11797E-11	-1.48362E-02	0.00000E+00	0.00000E+00	0.00000E+00	1.21414E-06
22	-2.18820E-11	-1.48377E-02	0.00000E+00	0.00000E+00	0.00000E+00	9.82647E-07

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 1
LOADTYPE DEAD TITLE DL

STAAD SPACE -- PAGE NO. 9

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.236850505E+01
Y = 0.842859851E+02
Z = -0.537499985E+00

TOTAL APPLIED LOAD 1

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -0.26
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -0.14 MY= 0.00 MZ= -0.61

TOTAL REACTION LOAD 1

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.26
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.14 MY= -0.00 MZ= 0.61

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 1)

	MAXIMUMS	AT NODE
X =	4.04397E-03	4
Y =	-7.90097E-02	5
Z =	0.00000E+00	0
RX=	0.00000E+00	0
RY=	0.00000E+00	0
RZ=	5.18432E-04	21

EXTERNAL AND INTERNAL JOINT LOAD SUMMARY (KN METE)-

JT	EXT FX/ INT FX	EXT FY/ INT FY	EXT FZ/ INT FZ	EXT MX/ INT MX	EXT MY/ INT MY	EXT MZ/ INT MZ	
							SUPPORT=1
1	0.00	-0.00	0.00	0.00	0.00	0.00	
	-0.16	-0.06	0.00	0.00	0.00	-0.00	111000
2	0.00	-0.00	0.00	0.00	0.00	0.00	
	0.16	-0.06	0.00	0.00	0.00	-0.00	111000
12	0.00	-0.00	0.00	0.00	0.00	0.00	
	-0.16	-0.06	0.00	0.00	0.00	-0.00	111000
13	0.00	-0.00	0.00	0.00	0.00	0.00	
	0.16	-0.06	0.00	0.00	0.00	0.00	111000

FOR LOADING - 2
APPLIED JOINT EQUIVALENT LOADS

JOINT	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
3	4.81575E-10	-1.03855E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.02174E-02
4	6.14355E-10	-1.03854E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.02174E-02
5	1.09065E-10	-2.07751E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.49348E-07

STAAD SPACE -- PAGE NO. 10

APPLIED JOINT EQUIVALENT LOADS

JOINT	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
6	-1.51699E-09	-2.07710E-01	0.00000E+00	0.00000E+00	0.00000E+00	-2.70547E-05
7	-7.08838E-10	-2.07706E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.70507E-05
8	1.29845E-09	-2.07725E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.33973E-05
9	-1.04596E-09	-2.07710E-01	0.00000E+00	0.00000E+00	0.00000E+00	2.71380E-05
10	4.33771E-10	-2.07707E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.69994E-05
11	-4.94457E-10	-2.07727E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.37576E-05
14	4.81575E-10	-1.03855E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.02174E-02
15	6.14355E-10	-1.03854E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.02174E-02
16	1.09065E-10	-2.07751E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.49348E-07
17	-1.51699E-09	-2.07710E-01	0.00000E+00	0.00000E+00	0.00000E+00	-2.70547E-05
18	-7.08838E-10	-2.07706E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.70507E-05
19	1.29845E-09	-2.07725E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.33973E-05
20	-1.04596E-09	-2.07710E-01	0.00000E+00	0.00000E+00	0.00000E+00	2.71380E-05
21	4.33771E-10	-2.07707E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.69994E-05
22	-4.94457E-10	-2.07727E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.37576E-05

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 2
LOADTYPE DEAD TITLE LL

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.236850500E+01
Y = 0.843019777E+02
Z = -0.537499985E+00

TOTAL APPLIED LOAD 2

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -3.32
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -1.79 MY= 0.00 MZ= -7.87

TOTAL REACTION LOAD 2

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 3.32
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 1.79 MY= -0.00 MZ= 7.87

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 2)

	MAXIMUMS	AT NODE
X =	5.66155E-02	4
Y =	-1.10612E+00	5
Z =	0.00000E+00	0
RX=	0.00000E+00	0
RY=	0.00000E+00	0
RZ=	7.25804E-03	21

STAAD SPACE -- PAGE NO. 11

EXTERNAL AND INTERNAL JOINT LOAD SUMMARY (KN METE)-

JT	EXT FX/ INT FX	EXT FY/ INT FY	EXT FZ/ INT FZ	EXT MX/ INT MX	EXT MY/ INT MY	EXT MZ/ INT MZ	
							SUPPORT=1
1	0.00	0.00	0.00	0.00	0.00	0.00	
	-2.21	-0.83	0.00	0.00	0.00	-0.00	111000
2	0.00	0.00	0.00	0.00	0.00	0.00	
	2.21	-0.83	0.00	0.00	0.00	0.00	111000
12	0.00	0.00	0.00	0.00	0.00	0.00	
	-2.21	-0.83	0.00	0.00	0.00	-0.00	111000
13	0.00	0.00	0.00	0.00	0.00	0.00	
	2.21	-0.83	0.00	0.00	0.00	0.00	111000

FOR LOADING - 3
APPLIED JOINT EQUIVALENT LOADS

JOINT	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
1	9.65095E-02	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
3	1.24635E-01	-1.48364E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.45963E-02
4	-1.89428E-09	-1.48364E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.45962E-02
5	4.13511E-03	-2.96787E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.00031E-07
6	4.82078E-02	-2.96729E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.86505E-05
7	3.20777E-02	-2.96724E-01	0.00000E+00	0.00000E+00	0.00000E+00	-2.43584E-05
8	1.61300E-02	-2.96751E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.91375E-05
9	1.33819E-09	-2.96729E-01	0.00000E+00	0.00000E+00	0.00000E+00	3.87684E-05
10	4.45952E-10	-2.96724E-01	0.00000E+00	0.00000E+00	0.00000E+00	2.42826E-05
11	4.48285E-10	-2.96753E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.96527E-05
12	9.65095E-02	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
14	1.24635E-01	-1.48364E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.45963E-02
15	-1.89428E-09	-1.48364E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.45962E-02
16	4.13511E-03	-2.96787E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.00031E-07
17	4.82078E-02	-2.96729E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.86505E-05
18	3.20777E-02	-2.96724E-01	0.00000E+00	0.00000E+00	0.00000E+00	-2.43584E-05
19	1.61300E-02	-2.96751E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.91375E-05
20	1.33819E-09	-2.96729E-01	0.00000E+00	0.00000E+00	0.00000E+00	3.87684E-05
21	4.45952E-10	-2.96724E-01	0.00000E+00	0.00000E+00	0.00000E+00	2.42826E-05
22	4.48285E-10	-2.96753E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.96527E-05

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 3
LOADTYPE NONE TITLE WL X

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.325780229E+00
Y = 0.841751991E+02
Z = -0.537499986E+00

STAAD SPACE -- PAGE NO. 12

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.236850500E+01
Y = 0.843019777E+02
Z = -0.537499985E+00

TOTAL APPLIED LOAD 3

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = 0.64
SUMMATION FORCE-Y = -4.75
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -2.55 MY= -0.35 MZ= -65.40

TOTAL REACTION LOAD 3

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = -0.64
SUMMATION FORCE-Y = 4.75
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 2.55 MY= 0.35 MZ= 65.40

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 3)

	MAXIMUMS	AT NODE
X =	9.82924E-02	4
Y =	-1.56811E+00	5
Z =	0.00000E+00	0
RX=	0.00000E+00	0
RY=	0.00000E+00	0
RZ=	1.03774E-02	21

EXTERNAL AND INTERNAL JOINT LOAD SUMMARY (KN METE)-

JT	EXT FX/ INT FX	EXT FY/ INT FY	EXT FZ/ INT FZ	EXT MX/ INT MX	EXT MY/ INT MY	EXT MZ/ INT MZ	
							SUPPORT=1
1	0.10	0.00	0.00	0.00	0.00	0.00	
	-3.02	-1.18	0.00	0.00	0.00	-0.00	111000
2	0.00	0.00	0.00	0.00	0.00	0.00	
	3.25	-1.20	0.00	0.00	0.00	-0.00	111000
12	0.10	0.00	0.00	0.00	0.00	0.00	
	-3.02	-1.18	0.00	0.00	0.00	-0.00	111000
13	0.00	0.00	0.00	0.00	0.00	0.00	
	3.25	-1.20	0.00	0.00	0.00	0.00	111000

FOR LOADING - 4
APPLIED JOINT EQUIVALENT LOADS

JOINT	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
3	1.07925E-09	-1.48364E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.45963E-02
4	-1.89428E-09	-1.48364E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.45962E-02
5	6.77376E-12	-2.96787E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.00031E-07

STAAD SPACE -- PAGE NO. 13

APPLIED JOINT EQUIVALENT LOADS

JOINT	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
6	2.67616E-09	-2.96729E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.86505E-05
7	-1.63052E-09	-2.96724E-01	0.00000E+00	0.00000E+00	0.00000E+00	-2.43584E-05
8	8.60346E-10	-2.96751E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.91375E-05
9	1.33819E-09	-2.96729E-01	0.00000E+00	0.00000E+00	0.00000E+00	3.87684E-05
10	4.45952E-10	-2.96724E-01	0.00000E+00	0.00000E+00	0.00000E+00	2.42826E-05
11	4.48285E-10	-2.96753E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.96527E-05
12	0.00000E+00	0.00000E+00	5.24101E-01	0.00000E+00	0.00000E+00	0.00000E+00
13	0.00000E+00	0.00000E+00	5.24101E-01	0.00000E+00	0.00000E+00	0.00000E+00
14	1.07925E-09	-1.48364E-01	2.88784E-01	0.00000E+00	0.00000E+00	-1.45963E-02
15	-1.89428E-09	-1.48364E-01	2.88784E-01	0.00000E+00	0.00000E+00	1.45962E-02
16	6.77376E-12	-2.96787E-01	9.95591E-02	0.00000E+00	0.00000E+00	-5.00031E-07
17	2.67616E-09	-2.96729E-01	1.39072E-01	0.00000E+00	0.00000E+00	-3.86505E-05
18	-1.63052E-09	-2.96724E-01	1.16991E-01	0.00000E+00	0.00000E+00	-2.43584E-05
19	8.60346E-10	-2.96751E-01	1.03847E-01	0.00000E+00	0.00000E+00	-1.91375E-05
20	1.33819E-09	-2.96729E-01	1.39072E-01	0.00000E+00	0.00000E+00	3.87684E-05
21	4.45952E-10	-2.96724E-01	1.16991E-01	0.00000E+00	0.00000E+00	2.42826E-05
22	4.48285E-10	-2.96753E-01	1.03847E-01	0.00000E+00	0.00000E+00	1.96527E-05

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 4
LOADTYPE NONE TITLE WL Z

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.236850500E+01
Y = 0.843019777E+02
Z = -0.537499985E+00

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.236850508E+01
Y = 0.841540033E+02
Z = -0.107499997E+01

TOTAL APPLIED LOAD 4

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -4.75
SUMMATION FORCE-Z = 2.45

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 203.22 MY= -5.79 MZ= -11.25

TOTAL REACTION LOAD 4

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 4.75
SUMMATION FORCE-Z = -2.45

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -202.84 MY= 5.79 MZ= 11.25

STAAD SPACE -- PAGE NO. 14

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 4)

	MAXIMUMS	AT NODE
X =	8.08793E-02	4
Y =	-1.58017E+00	16
Z =	3.92012E+05	5
RX=	1.12003E+04	12
RY=	1.85352E-02	2
RZ=	1.03686E-02	10

EXTERNAL AND INTERNAL JOINT LOAD SUMMARY (KN METE)-

JT	EXT FX/ INT FX	EXT FY/ INT FY	EXT FZ/ INT FZ	EXT MX/ INT MX	EXT MY/ INT MY	EXT MZ/ INT MZ	
							SUPPORT=1
1	0.00	0.00	0.00	0.00	0.00	0.00	
	-3.16	-1.19	0.35	-0.00	-0.00	0.00	111000
2	0.00	0.00	0.00	0.00	0.00	0.00	
	3.16	-1.19	1.24	0.38	-0.00	-0.00	111000
12	0.00	0.00	0.52	0.00	0.00	0.00	
	-3.16	-1.19	0.35	-0.00	0.00	-0.00	111000
13	0.00	0.00	0.52	0.00	0.00	0.00	
	3.16	-1.19	-0.55	0.00	0.00	0.00	111000

FOR LOADING - 5
APPLIED JOINT EQUIVALENT LOADS

JOINT	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
2	-9.65095E-02	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
3	1.07925E-09	-1.48364E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.45963E-02
4	-1.24635E-01	-1.48364E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.45962E-02
5	-4.13511E-03	-2.96787E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.00031E-07
6	2.67616E-09	-2.96729E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.86505E-05
7	-1.63052E-09	-2.96724E-01	0.00000E+00	0.00000E+00	0.00000E+00	-2.43584E-05
8	8.60346E-10	-2.96751E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.91375E-05
9	-4.82078E-02	-2.96729E-01	0.00000E+00	0.00000E+00	0.00000E+00	3.87684E-05
10	-3.20777E-02	-2.96724E-01	0.00000E+00	0.00000E+00	0.00000E+00	2.42826E-05
11	-1.61300E-02	-2.96753E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.96527E-05
13	-9.65095E-02	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
14	1.07925E-09	-1.48364E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.45963E-02
15	-1.24635E-01	-1.48364E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.45962E-02
16	-4.13511E-03	-2.96787E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.00031E-07
17	2.67616E-09	-2.96729E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.86505E-05
18	-1.63052E-09	-2.96724E-01	0.00000E+00	0.00000E+00	0.00000E+00	-2.43584E-05
19	8.60346E-10	-2.96751E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.91375E-05
20	-4.82078E-02	-2.96729E-01	0.00000E+00	0.00000E+00	0.00000E+00	3.87684E-05
21	-3.20777E-02	-2.96724E-01	0.00000E+00	0.00000E+00	0.00000E+00	2.42826E-05
22	-1.61300E-02	-2.96753E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.96527E-05

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 5
LOADTYPE NONE TITLE WL-X

STAAD SPACE -- PAGE NO. 15

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.441122981E+01
Y = 0.841751982E+02
Z = -0.537499981E+00

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.236850500E+01
Y = 0.843019777E+02
Z = -0.537499985E+00

TOTAL APPLIED LOAD 5

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 5)
SUMMATION FORCE-X = -0.64
SUMMATION FORCE-Y = -4.75
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -2.55 MY= 0.35 MZ= 42.91

TOTAL REACTION LOAD 5

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 5)
SUMMATION FORCE-X = 0.64
SUMMATION FORCE-Y = 4.75
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 2.55 MY= -0.35 MZ= -42.91

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 5)

	MAXIMUMS	AT NODE
X =	-9.82923E-02	14
Y =	-1.56811E+00	5
Z =	0.00000E+00	0
RX=	0.00000E+00	0
RY=	0.00000E+00	0
RZ=	-1.03774E-02	18

EXTERNAL AND INTERNAL JOINT LOAD SUMMARY (KN METE)-

JT	EXT FX/ INT FX	EXT FY/ INT FY	EXT FZ/ INT FZ	EXT MX/ INT MX	EXT MY/ INT MY	EXT MZ/ INT MZ	
							SUPPORT=1
1	0.00	0.00	0.00	0.00	0.00	0.00	
	-3.25	-1.20	0.00	0.00	0.00	0.00	111000
2	-0.10	0.00	0.00	0.00	0.00	0.00	
	3.02	-1.18	0.00	0.00	0.00	-0.00	111000
12	0.00	0.00	0.00	0.00	0.00	0.00	
	-3.25	-1.20	0.00	0.00	0.00	-0.00	111000
13	-0.10	0.00	0.00	0.00	0.00	0.00	
	3.02	-1.18	0.00	0.00	0.00	-0.00	111000

STAAD SPACE -- PAGE NO. 16

FOR LOADING - 6
APPLIED JOINT EQUIVALENT LOADS

JOINT	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
3	1.07925E-09	-1.48364E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.45963E-02
4	-1.89428E-09	-1.48364E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.45962E-02
5	6.77376E-12	-2.96787E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.00031E-07
6	2.67616E-09	-2.96729E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.86505E-05
7	-1.63052E-09	-2.96724E-01	0.00000E+00	0.00000E+00	0.00000E+00	-2.43584E-05
8	8.60346E-10	-2.96751E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.91375E-05
9	1.33819E-09	-2.96729E-01	0.00000E+00	0.00000E+00	0.00000E+00	3.87684E-05
10	4.45952E-10	-2.96724E-01	0.00000E+00	0.00000E+00	0.00000E+00	2.42826E-05
11	4.48285E-10	-2.96753E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.96527E-05
12	0.00000E+00	0.00000E+00	5.24101E-01	0.00000E+00	0.00000E+00	0.00000E+00
13	0.00000E+00	0.00000E+00	5.24101E-01	0.00000E+00	0.00000E+00	0.00000E+00
14	1.07925E-09	-1.48364E-01	2.88784E-01	0.00000E+00	0.00000E+00	-1.45963E-02
15	-1.89428E-09	-1.48364E-01	2.88784E-01	0.00000E+00	0.00000E+00	1.45962E-02
16	6.77376E-12	-2.96787E-01	9.95591E-02	0.00000E+00	0.00000E+00	-5.00031E-07
17	2.67616E-09	-2.96729E-01	1.39072E-01	0.00000E+00	0.00000E+00	-3.86505E-05
18	-1.63052E-09	-2.96724E-01	1.16991E-01	0.00000E+00	0.00000E+00	-2.43584E-05
19	8.60346E-10	-2.96751E-01	1.03847E-01	0.00000E+00	0.00000E+00	-1.91375E-05
20	1.33819E-09	-2.96729E-01	1.39072E-01	0.00000E+00	0.00000E+00	3.87684E-05
21	4.45952E-10	-2.96724E-01	1.16991E-01	0.00000E+00	0.00000E+00	2.42826E-05
22	4.48285E-10	-2.96753E-01	1.03847E-01	0.00000E+00	0.00000E+00	1.96527E-05

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 6
LOADTYPE NONE TITLE WL-Z

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.236850500E+01
Y = 0.843019777E+02
Z = -0.537499985E+00

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.236850508E+01
Y = 0.841540033E+02
Z = -0.107499997E+01

TOTAL APPLIED LOAD 6

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 6)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -4.75
SUMMATION FORCE-Z = 2.45

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 203.22 MY= -5.79 MZ= -11.25

TOTAL REACTION LOAD 6

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 6)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 4.75
SUMMATION FORCE-Z = -2.45

STAAD SPACE -- PAGE NO. 17

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -202.84 MY= 5.79 MZ= 11.25

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 6)

	MAXIMUMS	AT NODE
X =	8.08793E-02	4
Y =	-1.58017E+00	16
Z =	3.92012E+05	5
RX=	1.12003E+04	12
RY=	1.85352E-02	2
RZ=	1.03686E-02	10

EXTERNAL AND INTERNAL JOINT LOAD SUMMARY (KN METE)-

JT	EXT FX/ INT FX	EXT FY/ INT FY	EXT FZ/ INT FZ	EXT MX/ INT MX	EXT MY/ INT MY	EXT MZ/ INT MZ	
							SUPPORT=1
1	0.00	0.00	0.00	0.00	0.00	0.00	
	-3.16	-1.19	0.35	-0.00	-0.00	0.00	111000
2	0.00	0.00	0.00	0.00	0.00	0.00	
	3.16	-1.19	1.24	0.38	-0.00	-0.00	111000
12	0.00	0.00	0.52	0.00	0.00	0.00	
	-3.16	-1.19	0.35	-0.00	0.00	-0.00	111000
13	0.00	0.00	0.52	0.00	0.00	0.00	
	3.16	-1.19	-0.55	0.00	0.00	0.00	111000

LOAD COMBINATION NO. 101
1.5(DL+LL)

LOADING- 1. 2.
FACTOR - 1.50 1.50

LOAD COMBINATION NO. 102
1.5(DL+WLX)

LOADING- 1. 3.
FACTOR - 1.50 1.50

LOAD COMBINATION NO. 103
1.5(DL+WLZ)

LOADING- 1. 4.
FACTOR - 1.50 1.50

LOAD COMBINATION NO. 104
1.5(DL+WL-X)

LOADING- 1. 5.
FACTOR - 1.50 1.50

LOAD COMBINATION NO. 105
1.5(DL+WL-Z)

```
LOADING-     1.     6.
FACTOR -   1.50   1.50

  LOAD COMBINATION NO.   201
  DL + LL

LOADING-     1.     2.
FACTOR -   1.00   1.00

  LOAD COMBINATION NO.   202
  DL+WLX

LOADING-     1.     3.
FACTOR -   1.00   1.00

  LOAD COMBINATION NO.   203
  DL+WLZ

LOADING-     4.     1.
FACTOR -   1.00   1.00

  LOAD COMBINATION NO.   204
  DL+WL-X

LOADING-     1.     5.
FACTOR -   1.00   1.00

  LOAD COMBINATION NO.   205
  DL+WL-Z

LOADING-     6.     1.
FACTOR -   1.00   1.00

  LOAD COMBINATION NO.   206
  DL+0.8WLX

LOADING-     1.     3.
FACTOR -   1.00   0.80

  LOAD COMBINATION NO.   207
  DL+0.8WLZ

LOADING-     4.     1.
FACTOR -   0.80   1.00

  LOAD COMBINATION NO.   208
  DL+0.8WL-X

LOADING-     5.     1.
FACTOR -   0.80   1.00

  LOAD COMBINATION NO.   209
  DL+0.8WL-Z
```

STAAD SPACE -- PAGE NO. 19

LOADING- 6. 1.
FACTOR - 0.80 1.00

********** END OF DATA FROM INTERNAL STORAGE **********

102. PARAMETER 1
103. CODE IS800 LSD
104. FYLD 214000 ALL
105. CHECK CODE ALL
STEEL DESIGN

Member Number: 1					
Member Section: ST 60X60X3.2SHS (APL APOLLOTUBE SECTIONS)					
Status: PASS	Ratio: 0.375	Critical Load Case: 104	Location:	0.21	
Critical Condition: Sec. 8.2.1.2					

Member Number: 2					
Member Section: ST 60X60X3.2SHS (APL APOLLOTUBE SECTIONS)					
Status: PASS	Ratio: 0.375	Critical Load Case: 102	Location:	0.21	
Critical Condition: Sec. 8.2.1.2					

Member Number: 3					
Member Section: ST 60X60X3.2SHS (APL APOLLOTUBE SECTIONS)					
Status: PASS	Ratio: 0.375	Critical Load Case: 104	Location:	0.00	
Critical Condition: Sec. 8.2.1.2					

Member Number: 4					
Member Section: ST 60X60X3.2SHS (APL APOLLOTUBE SECTIONS)					
Status: PASS	Ratio: 0.173	Critical Load Case: 103	Location:	0.00	
Critical Condition: Sec. 9.3.2.2					

STAAD SPACE -- PAGE NO. 20

Member Number: 5					
Member Section: ST 60X60X3.2SHS (APL APOLLOTUBE SECTIONS)					
Status: PASS	Ratio: 0.230	Critical Load Case: 103	Location:	0.59	
Critical Condition: Sec. 9.3.2.2					

Member Number: 6					
Member Section: ST 60X60X3.2SHS (APL APOLLOTUBE SECTIONS)					
Status: PASS	Ratio: 0.253	Critical Load Case: 103	Location:	0.59	
Critical Condition: Sec. 9.3.2.2					

Member Number: 7					
Member Section: ST 60X60X3.2SHS (APL APOLLOTUBE SECTIONS)					
Status: PASS	Ratio: 0.375	Critical Load Case: 102	Location:	0.00	
Critical Condition: Sec. 8.2.1.2					

Member Number: 8					
Member Section: ST 60X60X3.2SHS (APL APOLLOTUBE SECTIONS)					
Status: PASS	Ratio: 0.169	Critical Load Case: 103	Location:	0.00	
Critical Condition: Sec. 9.3.2.2					

Member Number: 9					
Member Section: ST 60X60X3.2SHS (APL APOLLOTUBE SECTIONS)					
Status: PASS	Ratio: 0.226	Critical Load Case: 103	Location:	0.59	
Critical Condition: Sec. 9.3.2.2					

STAAD.PRO CODE CHECKING - IS-800-2007-LSD (V2.1)
****

Member Number:	10				
Member Section:	ST	60X60X3.2SHS	(APL APOLLOTUBE SECTIONS)		
Status: PASS	Ratio:	0.252	Critical Load Case: 103	Location:	0.59
Critical Condition: Sec. 9.3.2.2					

Member Number:	13				
Member Section:	ST	25X25X2.0SHS	(APL APOLLOTUBE SECTIONS)		
Status: PASS	Ratio:	0.030	Critical Load Case: 103	Location:	0.00
Critical Condition: Sec. 7.1.2					

Member Number:	14				
Member Section:	ST	25X25X2.0SHS	(APL APOLLOTUBE SECTIONS)		
Status: PASS	Ratio:	0.218	Critical Load Case: 103	Location:	0.00
Critical Condition: Sec. 7.1.2					

Member Number:	15				
Member Section:	ST	25X25X2.0SHS	(APL APOLLOTUBE SECTIONS)		
Status: PASS	Ratio:	0.010	Critical Load Case: 103	Location:	0.00
Critical Condition: Sec. 7.1.2					

STAAD.PRO CODE CHECKING - IS-800-2007-LSD (V2.1)
****

Member Number:	16				
Member Section:	ST 25X25X2.0SHS	(APL APOLLOTUBE SECTIONS)			
Status: PASS	Ratio: 0.015	Critical Load Case:	103	Location:	0.00
Critical Condition: Sec. 7.1.2					

Member Number:	17				
Member Section:	ST 25X25X2.0SHS	(APL APOLLOTUBE SECTIONS)			
Status: PASS	Ratio: 0.012	Critical Load Case:	103	Location:	0.00
Critical Condition: Sec. 7.1.2					

Member Number:	18				
Member Section:	ST 25X25X2.0SHS	(APL APOLLOTUBE SECTIONS)			
Status: PASS	Ratio: 0.011	Critical Load Case:	103	Location:	0.00
Critical Condition: Sec. 7.1.2					

Member Number:	19				
Member Section:	ST 25X25X2.0SHS	(APL APOLLOTUBE SECTIONS)			
Status: PASS	Ratio: 0.018	Critical Load Case:	103	Location:	0.00
Critical Condition: Sec. 7.1.2					

STAAD SPACE -- PAGE NO. 23

STAAD.PRO CODE CHECKING - IS-800-2007-LSD (V2.1)
****

Member Number: 20					
Member Section: ST 25X25X2.0SHS (APL APOLLOTUBE SECTIONS)					
Status: PASS	Ratio: 0.010	Critical Load Case:	103	Location:	0.00
Critical Condition: Sec. 7.1.2					

Member Number: 21					
Member Section: ST 25X25X2.0SHS (APL APOLLOTUBE SECTIONS)					
Status: PASS	Ratio: 0.011	Critical Load Case:	103	Location:	0.00
Critical Condition: Sec. 7.1.2					

Member Number: 22					
Member Section: ST 60X60X3.2SHS (APL APOLLOTUBE SECTIONS)					
Status: PASS	Ratio: 0.375	Critical Load Case:	104	Location:	0.21
Critical Condition: Sec. 8.2.1.2					

Member Number: 23					
Member Section: ST 60X60X3.2SHS (APL APOLLOTUBE SECTIONS)					
Status: PASS	Ratio: 0.375	Critical Load Case:	102	Location:	0.21
Critical Condition: Sec. 8.2.1.2					

STAAD SPACE -- PAGE NO. 24

STAAD.PRO CODE CHECKING - IS-800-2007-LSD (V2.1)
****

Member Number:	24				
Member Section:	ST 60X60X3.2SHS	(APL APOLLOTUBE SECTIONS)			
Status: PASS	Ratio: 0.375	Critical Load Case:	104	Location:	0.00
Critical Condition: Sec. 8.2.1.2					

Member Number:	25				
Member Section:	ST 60X60X3.2SHS	(APL APOLLOTUBE SECTIONS)			
Status: PASS	Ratio: 0.173	Critical Load Case:	103	Location:	0.00
Critical Condition: Sec. 9.3.2.2					

Member Number:	26				
Member Section:	ST 60X60X3.2SHS	(APL APOLLOTUBE SECTIONS)			
Status: PASS	Ratio: 0.230	Critical Load Case:	103	Location:	0.59
Critical Condition: Sec. 9.3.2.2					

Member Number:	27				
Member Section:	ST 60X60X3.2SHS	(APL APOLLOTUBE SECTIONS)			
Status: PASS	Ratio: 0.253	Critical Load Case:	103	Location:	0.59
Critical Condition: Sec. 9.3.2.2					

STAAD SPACE -- PAGE NO. 25

STAAD.PRO CODE CHECKING - IS-800-2007-LSD (V2.1)
****

Member Number: 28					
Member Section: ST 60X60X3.2SHS (APL APOLLOTUBE SECTIONS)					
Status: PASS	Ratio: 0.375	Critical Load Case:	102	Location:	0.00
Critical Condition: Sec. 8.2.1.2					

Member Number: 29					
Member Section: ST 60X60X3.2SHS (APL APOLLOTUBE SECTIONS)					
Status: PASS	Ratio: 0.164	Critical Load Case:	103	Location:	0.00
Critical Condition: Sec. 9.3.2.2					

Member Number: 30					
Member Section: ST 60X60X3.2SHS (APL APOLLOTUBE SECTIONS)					
Status: PASS	Ratio: 0.226	Critical Load Case:	103	Location:	0.59
Critical Condition: Sec. 9.3.2.2					

Member Number: 31					
Member Section: ST 60X60X3.2SHS (APL APOLLOTUBE SECTIONS)					
Status: PASS	Ratio: 0.252	Critical Load Case:	103	Location:	0.59
Critical Condition: Sec. 9.3.2.2					

106. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 3,2023 TIME= 17:42:16 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://www.bentley.com/en/support/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2017 Bentley Systems, Inc. *
* http://www.bentley.com *
****